

Oblique surface deflector

As per newton's second law of motion, the force of the jet strikes a plate is equal to the rate of change of momentum of the jet, therefore; For the oblique deflector ($\theta = 30^\circ, 45^\circ$ or 60°):

$$F = \text{Mass of water flowing} \times \text{Change of velocity}$$

So,

$$F = \rho AV \times \Delta V = \rho AV \times (V \sin(\theta) - 0)$$

And since,

$$F_x = F \sin(\theta) = \rho AV^2 \sin(\theta) \times \sin(\theta)$$

Therefore:

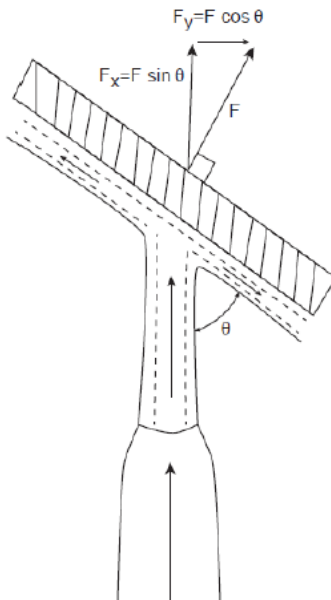
$$F_x = \rho AV^2 (\sin(\theta))^2$$

where:

$$V^2 = V_o^2 - 2gS$$

and,

$$V_o = Q/A$$



Where:

F = the force exerted by the jet in N.

ρ = density of water in kg/m³.

A = cross sectional area of the jet in m².

V = actual velocity at the center of the jet in m/s.

ΔV = change of the velocity in m/s.

V_0 = velocity of the jet in m/s.

S = the distant between the nozzle opening and the deflector opening in m.

Q = the flow rate in m³/s.

Answer:

$A = 0.0000785 \text{ m}^2$, $S = 0.035 \text{ m}$, $g = 9.81 \text{ m/s}^2$, $\rho = 1000 \text{ kg/m}^3$

❖ $Q = \text{-----} \text{ m}^3/\text{s}$.

❖ $V_o = Q / A = \text{-----} = \text{-----} \text{ m/s}$.

❖ $V^2 = V_o^2 - 2gS = \text{-----} = \text{-----} = \text{m/s}$.

❖ $\Theta = \text{-----}^\circ$.

❖ $F_{th} = \rho A V^2 (\sin(\Theta))^2 = \text{-----} = \text{-----} \text{ kg.m/s}^2 \text{ (N)}$

❖ **Fac.** = total weight = total mass $\times$ g = $\text{-----} \text{ kg.m/s}^2 \text{ (N)}$

❖ Plot the graph between F_{th} & F_{ac} . vs. Q :

Q (L/min.)									
F (theoretical)									
F (actual)									

